Worksheet # 23

Week 12, Day 1: Intro to network analysis

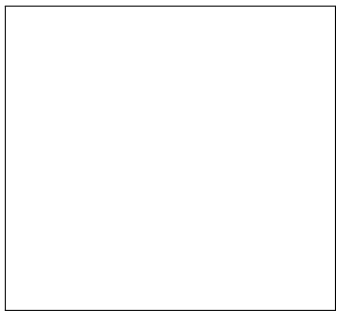
Name: _____
NetID: _____
Date: _____

Guided checkpoints

For full credit, fill in the checkpoints. If you skip a checkpoint, include your own notes for full credit.

1. Bridges of Königsberg

Draw the graph: one node per land mass, one edge per bridge. Fill in the adjacency matrix (bridges between each pair) and each degree.

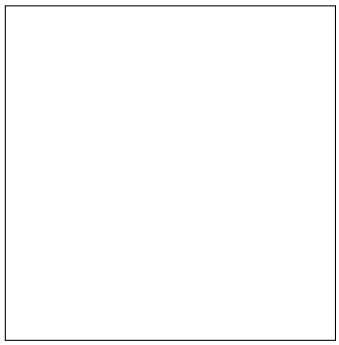


	A	K	L	V	deg
A					
K					
L					
V					

Odd-degree land masses: _____

2. Bridges of Shikaakwa

Draw the graph for the bridges on the slide (G Goose Island, L Loop, N North Side, W West Side). Fill in the adjacency matrix and each degree.



	G	L	N	W	deg
G					
L					
N					
W					

Can one walk cross every bridge exactly once?

What rule(s) did Euler find for when a walk can cross every bridge exactly once?

Open-ended notes

3. Vocab

Write a short definition and/or example for each.

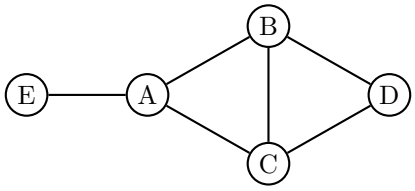
Degree
Self-loop
Directed graph
In-degree
Out-degree
Weighted graph
Complete bipartite graph

4. Complete Bipartite Graph

Draw the complete bipartite graph $K_{2,3}$.

5. Triadic Closure

In this friendship graph, circle a missing edge most likely to form next. Explain your choice in terms of shared friends.



6. Your Project as a Network

Try to draw a graph from your project's data/concepts.

A node is... An edge means...